

# Midterm Exam 1 · Fall 2022

MATH 411 · October 7, 2022

## Complete worked solutions

Retypeset past exam for practice. The original allowed one handwritten page of notes, lasted 50 minutes, and was worth 25 points.

1. (a) **(3 points)** Find every solution of  $z^3 = -8i$ . You may leave your answers in polar form  $re^{i\theta}$ .

**Solution.** Write  $z = re^{i\theta}$ , so  $z^3 = -8i = r^3 e^{3i\theta}$ . We have

$$-8i = 8(-i) = 8e^{i(3\pi/2)}, \quad r^3 e^{i(3\theta)} = 8e^{i(3\pi/2)}.$$

Thus  $r = 2$  and, modulo  $2\pi$ , the three values of  $\theta$  are

$$\begin{aligned} \theta &= \frac{\pi}{2}, \\ \theta &= \frac{\pi}{2} + \frac{2\pi}{3} = \frac{7\pi}{6}, \\ \theta &= \frac{\pi}{2} + \frac{2\pi \cdot 2}{3} = \frac{11\pi}{6}. \end{aligned}$$

- (b) **(4 points)** Let  $r > 0$  be real and  $w \in \mathbb{C}$ . Show that

$$\left| \frac{w-r}{1-r\bar{w}} \right| \leq 1 \quad \text{when } r < 1 \text{ and } |w| < 1.$$

**Solution.** The denominator is nonzero because  $|\bar{w}r| < 1$ . Since  $r \in \mathbb{R}$ , we have  $\bar{r} = r$ , and

$$\begin{aligned} \left| \frac{w-r}{1-\bar{w}r} \right| \leq 1 &\iff |w-r| \leq |1-\bar{w}r| \\ &\iff |w-r|^2 \leq |1-\bar{w}r|^2 \\ &\iff (w-r)(\bar{w}-r) \leq (1-\bar{w}r)(1-wr) \\ &\iff w\bar{w} - r\bar{w} - rw + r^2 \leq 1 - \bar{w}r - wr + w\bar{w}r^2 \\ &\iff |w|^2 + r^2 \leq 1 + |w|^2 r^2. \end{aligned}$$

This is true as  $|w|^2 < 1$  and  $r^2 < 1$ : indeed,

$$1 + |w|^2 r^2 - (|w|^2 + r^2) = (1 - r^2)(1 - |w|^2) > 0.$$

In fact one has the strict inequality.

**2. (9 points)** Let  $D = \{z \in \mathbb{C} : |z| < 1\}$  and let  $f = u + iv$  be holomorphic on  $D$ . Suppose  $|f|^2 = u^2 + v^2 = c$  throughout  $D$ , where  $c > 0$ . Prove that  $f$  is constant.

*Hint.* Differentiate  $u^2 + v^2 = c$  with respect to  $x$  and  $y$ . You may use that a real function with continuous partial derivatives and both partial derivatives zero on  $D$  is constant.

**Solution.** Suppose  $|f|^2 = c$  with  $c > 0$ . Differentiating the expression  $|f|^2 = c$  yields the two expressions

$$2uu_x + 2vv_x = 0,$$

$$2uu_y + 2vv_y = 0.$$

Applying Cauchy–Riemann (i.e.  $u_x = v_y$ ,  $u_y = -v_x$ ) we find

$$2uu_x - 2vv_y = 0,$$

$$2uu_y + 2vv_x = 0.$$

Since

$$\det \begin{pmatrix} u & -v \\ v & u \end{pmatrix} = u^2 + v^2 \neq 0,$$

the only solution to this system is when  $u_x = u_y = 0$ . In a similar way we also conclude that  $v_x = v_y = 0$ . As the Jacobian matrix of the map  $F = (u(x, y), v(x, y))$  vanishes, the map  $F$  is locally constant, and hence constant since  $D$  is connected.

3. (9 points) Let  $f$  be entire and analytic, and set

$$A_n = \{z \in \mathbb{C} : f^{(n)}(z) = 0\}, \quad n \geq 0.$$

- (a) Show that  $A_n$  is closed. Show also that  $A_n$  is discrete if  $f^{(n)}$  is not identically zero. What is  $A_n$  if  $f^{(n)}$  is identically zero?

Here *discrete* means that every point of the set has a neighborhood containing no other point of the set.

**Solution.**  $A_n$  is closed and discrete if  $f$  is analytic and  $f^{(n)} \not\equiv 0$ . Otherwise  $A_n = \mathbb{C}$ , which is closed, but not discrete.

$A_n$  closed:  $f^{(n)}$  is continuous. Let  $z \notin A_n$ , say  $f^{(n)}(z) = \varepsilon \neq 0$ . Pick a small neighborhood around  $z$  such that

$$|f^{(n)}(z) - f^{(n)}(z')| < \frac{|\varepsilon|}{2} \quad \text{for } |z - z'| < \delta.$$

Then all  $z' \in B(z, \delta)$  satisfy  $|f^{(n)}(z')| > 0$ . Thus the complement of  $A_n$  is open, so  $A_n$  is closed.

$A_n$  discrete: As  $f$  is analytic, so is  $f^{(n)}$ , and therefore if  $f^{(n)} \not\equiv 0$ , it is given by a non-zero power series. If  $z_0 \in \mathbb{C}$  is such that  $f^{(n)}(z_0) = 0$  (i.e.  $z_0 \in A_n$ ), then by the isolated zeros theorem from class there is a small neighborhood around  $z_0$  such that  $f^{(n)}(z) \neq 0$  on this neighborhood except at  $z_0$ . So this neighborhood meets  $A_n$  only at  $z_0$ .

- (b) Suppose that for every  $a \in \mathbb{C}$ , at least one coefficient in the Taylor expansion

$$f(z) = \sum_{n \geq 0} c_n(a)(z - a)^n$$

is zero. Show that  $f$  is a polynomial. You may use part (a), the fact that closed discrete subsets of  $\mathbb{C}$  are countable, and the fact that  $\mathbb{C}$  is uncountable.

**Solution.** If  $f$  is not a polynomial,  $f^{(n)}(z)$  is non-zero (as a function) for all  $n \geq 0$ , so each  $A_n$  is countable by part (a). Indeed, each closed disc meets a closed discrete set in only finitely many points, since otherwise there would be an accumulation point in the set. Taking the union over discs of positive integer radius centered at 0 proves countability. Therefore

$$\bigcup_{n \geq 0} A_n \text{ is countable.}$$

Note:

$$\bigcup_{n \geq 0} A_n = \{a \in \mathbb{C} : f^{(n)}(a) = 0 \text{ for some } n \geq 0\}.$$

Now if

$$f(z) = \sum_{n \geq 0} c_n(z - a)^n, \quad \text{then} \quad c_n = \frac{f^{(n)}(a)}{n!}.$$

So our assumption implies

$$\mathbb{C} \subseteq \bigcup_{n \geq 0} A_n,$$

which contradicts countability unless  $f^{(n)} \equiv 0$  for some  $n$ , i.e.  $f$  is a polynomial.